

ModelSEED Structure Cleanup PubChem Cross-Validation Results

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Goal: Validate and correct the 30,321 compound structures in ModelSEED against PubChem.

Approach:

- 1 **Look up** each compound in PubChem via cross-references (ChEBI, KEGG), compound names, and direct structure lookups (InChIKey, InChI)
- 2 **Classify** discrepancies by comparing InChIKey blocks:
Block 1 (connectivity) – Block 2 (stereo) – Block 3 (protonation)
- 3 **Correct** when validation criteria are met: stereochemistry from PubChem (stereo count + inversion check), protonation to pH 7 (OpenBabel + pK_a check)
- 4 **Preserve** existing ionization states — do not replace with PubChem's neutral forms

Validation Results (30,321 compounds)

Category	Count	%
MATCH	19,984	65.9
PROT. DIFF	4,365	14.4
XREF_CONFLICT	2,919	9.6
MISMATCH	1,331	4.4
NOT_FOUND	832	2.7
STEREO_DIFF	683	2.3
Other	207	0.7
Total	30,321	

- **80.3%** share the same connectivity and stereochemistry with PubChem
- **2.3%** STEREO_DIFF — same molecule, missing stereocenters → correctable
- **4.4%** MISMATCH — different molecular graph → manual review
- **9.6%** xref conflicts — ChEBI $\neq$ KEGG mapping

Corrections Applied (201 compounds)

Type	Candidates	Accepted	Rejected	Criteria
Stereochemistry	683	188	495	PubChem $\geq$ stored stereocenters, no inversions, pH 7 normalized
pH 7 protonation	30,321	13	1	OpenBabel + pK_a validation

Stereo corrections (188):

- PubChem structure must define $\geq$ stereocenters than stored
- Shared stereocenters verified via InChI layer comparison (no inversions)
- Each correction pH 7-normalized before application (prevents PubChem's neutral protonation from overwriting curated charges)

pH 7 corrections (13): terminal phosphate/sulfate deprotonation. Free pyrophosphate excluded ($pK_{a4}=9.4 > \text{pH } 7$).

pH 7 Protonation Corrections (13 accepted, 1 rejected)

Compound	Name	Old	New
cpd00002	ATP	-3	-4
cpd00038	GTP	-3	-4
cpd00052	CTP	-3	-4
cpd00062	UTP	-3	-4
cpd00068	ITP	-3	-4
cpd00008	ADP	-2	-3
cpd00031	GDP	-2	-3
cpd00090	IDP	-2	-3
cpd00096	CDP	-2	-3
cpd00014	UDP	-2	-3
cpd00056	ThDP	-1	-2
cpd00103	PRPP	-4	-5
cpd00081	Sulfite	-1	-2
<i>Rejected:</i>			
cpd00012	PPi	-3	—

Accepted (13): each deprotonates one terminal **ester-linked** phosphate or sulfate group ($pK_a \approx 6.5$, below pH 7).

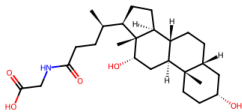
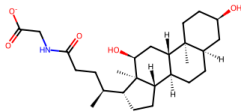
Rejected (1): free pyrophosphate (PPi). Its 4th pK_a is 9.4 — well above pH 7, so the proton remains. The pipeline detects this via a SMARTS-based check: if **no** phosphorus in the molecule has a C–O–P ester bond, the terminal pK_a is ~ 9 rather than ~ 6.5 .

cpd02877 — Glycodeoxycholate

cpd02877 [STEREO_DIFF]
ModelSEED

CORRECTED

PubChem



Formula: C₂₆H₄₂N₀S CID: 3035026 Strategy: name_lookup Query: 3alpha,12alpha-Dihydroxy-5beta-cholan
MS: C[C@@H](CCC(=O)NCC(=O)[O-])[C@H]1CC[C@@H]2[C@@H]3CC[C@@H]4C[C@H]1

cpd02877 (Glycodeoxycholate, name lookup). PubChem defines all stereocenters for the bile acid scaffold.

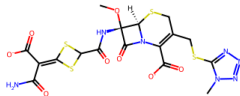
Accepted — no existing stereocenters inverted.

cpd04236 — Cefotetan

cpd04236 [STEREO_DIFF]
ModelSEED

CORRECTED

PubChem



Formula: C₁₇H₁₅N₇O₈S₄ CID: 53025 Strategy: kegg_xref Query: C06886
MS: CO[C@@H]1NC(=O)C2SC(=C(C(N)=O)C(=O)[O-])S2)C(=O)N2C(C(=O)[O-])

cpd04236 (Cefotetan, KEGG xref). PubChem defines additional stereocenters on the cephalosporin ring.

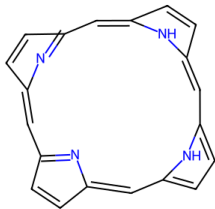
Accepted — adds missing stereochemistry.

Stereochemistry Correction Examples (cont.)

cpd03042 — Porphyrin

cpd03042 [STEREO_DIFF]

ModelSEED

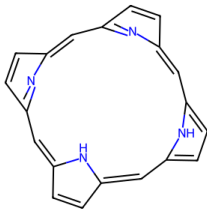


Formula: C₂₀H₁₄N₄ CID: 66868 Strategy: kegg_xref
MS: C1=Cc2cc3ccc(cc4ccc(cc5nc(cc1n2)C=C5)[nH]4)[nH]3

Query: C05113

CORRECTED

PubChem

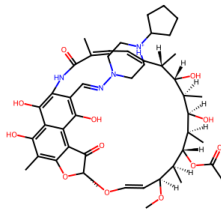


cpd03042 (Porphyrin, KEGG xref). PubChem replaces the UHFFFAOYSA achiral placeholder with a defined stereo block. **Accepted.**

cpd05038 — Rifapentine

cpd05038 [STEREO_DIFF]

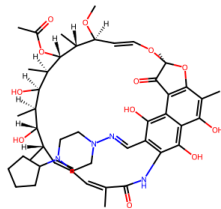
ModelSEED



Formula: C₄₇H₆₅N₄O₁₂ CID: 135403821 Strategy: kegg_xref Query: C08059
MS: CO[C@H]1/C=C/O[C@@]2(C)Oc3c(C)c(O)c4c(O)c(c(/C=N/N5CC[NH+]1(C...

CORRECTED

PubChem



cpd05038 (Rifapentine, KEGG xref). Macrolide antibiotic; PubChem record defines additional stereocenters. **Accepted.**

2,228 compounds were initially classified as mismatches or not-found via cross-references and name lookups.

Direct InChIKey lookup in PubChem recovered them — **97% returned exact InChIKey matches**.

Interpretation: for these compounds, the initial discrepancies were due to **incorrect cross-reference mappings or name ambiguity**, not incorrect stored structures.

Recovery result	Count
Exact match (MATCH)	2,161 (97.0%)
Stereo difference	51 (2.3%)
Still mismatched	2 (0.1%)
Other	14 (0.6%)

- 1 **XREF conflicts** (2,919) — ChEBI and KEGG map to different PubChem CIDs; need resolution strategy
- 2 **MISMATCH review** (1,331) — sub-classified into tautomers, wrong mappings, and structurally distinct compounds to prioritize manual review
- 3 **NOT_FOUND** (832) — compounds without PubChem records; mostly rare metabolites or generic class names
- 4 **AMBIGUOUS** (152) — compound names map to multiple PubChem CIDs

- **Single reference database** — validation relies on PubChem only; cross-checking against ChEBI, HMDB, or MetaCyc would provide independent corroboration
- **OpenBabel pH model** — empirical pK_a estimation; caught the free pyrophosphate error ($pK_{a4}=9.4$), but other edge cases may exist
- **Name-based lookups** — inherently noisy; multi-tier disambiguation (InChIKey > fingerprint similarity) mitigates but does not eliminate errors
- **No confidence scores** — binary accept/reject decisions without uncertainty quantification for borderline cases